WORKED SOLUTION

STEP II 2000, Question 13

Probability and Statistics • Capture-Recapture Estimation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 13.2, *Counting Voles*: https://beyondstep.keithmcnulty.org/ch13/voles.html

The key idea

This is capture–recapture: the second sample is a random choice of 200 voles from N, of which 200 are marked, so the number of marked voles in it has a hypergeometric distribution. To find where p_N is largest, don't differentiate (factorials of N aren't differentiable in any easy sense); instead compare consecutive terms through the ratio p_N/p_{N-1}, where almost everything cancels. The final identity then comes free: the third-sample probabilities must add up to 1.

Solution

The probability p_N

Model the second sample as a choice of 200 of the N voles, every subset of size 200 being equally likely. There are $\binom{N}{200}$ such subsets. Of the N voles, 200 are marked and $N-200$ are not. A sample with exactly 11 marked voles is a choice of 11 marked ones and 189 unmarked ones, so

$$p_N = \frac{\binom{200}{11}\binom{N-200}{189}}{\binom{N}{200}}.$$

Writing the two binomial coefficients that involve N as factorials,

$$p_N = \binom{200}{11} \cdot \frac{(N-200)!}{189!\,(N-389)!} \cdot \frac{200!\,(N-200)!}{N!}$$
$$= k\,\frac{((N-200)!)^2}{N!\,(N-389)!}, \qquad \text{where } k = \binom{200}{11}\frac{200!}{189!},$$

which does not depend on N. This is the required form. (It needs $N \geq 389$, since the two samples together contain $200 + 189 = 389$ different voles.)

Maximising p_N

For $N \geq 390$, divide p_N by p_{N-1}. Using $(N-200)! = (N-200)\,(N-201)!$, $N! = N\,(N-1)!$ and $(N-389)! = (N-389)\,(N-390)!$, everything cancels except

$$\frac{p_N}{p_{N-1}} = \frac{(N-200)^2}{N(N-389)}.$$

Both the numerator and the denominator are positive, so

$$\begin{aligned} p_N > p_{N-1} &\iff (N-200)^2 > N(N-389) \\ &\iff N^2 - 400N + 40000 > N^2 - 389N \\ &\iff 40000 > 11N \\ &\iff N < 3636.\overline{36}. \end{aligned}$$

Ask yourself What does this tell us about the shape of the sequence $p_{389}, p_{390}, \ldots$?

For $390 \leq N \leq 3636$ the ratio is greater than 1, so the sequence strictly increases up to p_{3636}. For $N \geq 3637$ the ratio is less than 1 (equality would need $11N = 40000$, which has no integer solution), so the sequence strictly decreases from then on. Hence p_N has a unique maximum, at

$$N = 3636.$$

This is the integer part of $\frac{200\times200}{11}$. It is what you'd guess by saying that the proportion marked in the sample, $\frac{11}{200}$, should match the proportion marked in the population, $\frac{200}{N}$.

The third sample

Take the population to be 3636, the estimate just found. After the second release, the marked voles are the original 200 plus the 189 newly marked ones, so 389 are marked and $3636 - 389 = 3247$ are unmarked. Reasoning exactly as before, the probability that a random sample of 200 contains exactly j marked voles is

$$\frac{\binom{389}{j}\binom{3247}{200-j}}{\binom{3636}{200}}, \qquad j = 0, 1, \ldots, 200.$$

Since $389 \geq 200$ and $3247 \geq 200$, every j from 0 to 200 is possible, and these are all the possible outcomes. They are mutually exclusive, so their probabilities add up to 1:

$$\sum_{j=0}^{200} \frac{\binom{389}{j}\binom{3247}{200-j}}{\binom{3636}{200}} = 1.$$

Multiplying by $\binom{3636}{200}$ gives

$$\sum_{j=0}^{200} \binom{389}{j}\binom{3247}{200-j} = \binom{3636}{200},$$

as required.

Checks

- Compute p_N exactly with integer binomial coefficients: it agrees with the factorial formula for several values of N (for example 389, 1000 and 5000), and a search over $389 \leq N < 20000$ puts the maximum at $N = 3636$.
- At the turning point, $p_{3636}/p_{3635} = 3436^2/(3636 \times 3246) \approx 1.0003 > 1$, but $p_{3637}/p_{3636} < 1$.
- The two sides of the final identity, computed exactly as integers, are equal. Note that $389 + 3247 = 3636$, as it must be.

Common slips

- Treating N as continuous and differentiating. Use the ratio of consecutive terms instead, and decide the inequality exactly.
- Forgetting that after the second release 389 voles, not 200, are marked.
- Saying "maximum at $N = 3636.36$": the population size must be a whole number, and you should check which neighbour wins.

Beyond STEP

Choosing the value of the parameter that makes the observed data most probable is the method of *maximum likelihood*, the central idea of first-year statistics. Here it gives the Lincoln–Petersen estimator for capture–recapture. The final identity is Vandermonde's identity, and the argument used, counting the same thing in two ways, is a powerful technique in combinatorics. Grimmett and Welsh, *Probability: An Introduction*, covers the hypergeometric distribution and this style of argument.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.